

Mitigation and Utilization of Methane Emissions from Landfills of Dhaka

Submitted by

Murmamirr Mohaimen

1910036

Mohammad Abdul Mohaimin

1910038

Ayan Chowdhury

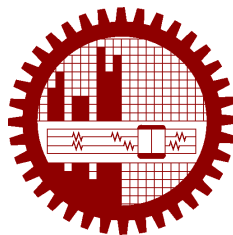
1910044

Md. Naymurr Rahman Sakib

1910047

Amitav Mandal

1910062



Department of Mechanical Engineering
Bangladesh University of Engineering and Technology

Dhaka, Bangladesh

Contents

List of Figures	ii
List of Tables	iii
<i>ABSTRACT</i>	iv
1 Introduction	1
2 Methane Emission Modeling and Estimation	4
2.1 Landfill Gas (LFG) Composition and Potential	4
2.1.1 Methane Estimation	4
2.2 Modelling Approaches	4
2.3 Calculations	5
2.4 Methane Production Capacity (L_o)	5
3 Mitigation and Utilization Methods for Landfill Gas (LFG)	7
3.1 Landfill Gas (LFG) Extraction Systems and Flaring Overview	7
3.2 Anaerobic Digester	11
3.2.1 Anaerobic Digestion of Aminbazar and Matuail Landfills	12
3.3 Incineration	14
References	17

List of Figures

3.1	LFG Recovery Plant System.	8
3.2	Vertical LFG Extraction Well	9
3.3	Horizontal LFG Extraction System	10
3.4	Extraction System with Each Well Connected to a Main Collection Pipe	10
3.5	Extraction System with Each Well Connected to a Pump and Regulation House	11
3.6	Anaerobic Digestion Process	12
3.7	Enter Caption	15

List of Tables

2.1	Methane Emission Estimation Over Time	6
3.1	Comparison of Anaerobic Digestion Parameters for Aminbazar and Matuail Landfills	14
3.2	Calorific Value and Composition of Different Waste Types at 70% Moisture Content	16
3.3	Energy Input and Power Output at Different Sites	16

ABSTRACT

Methane emissions from landfills pose a significant environmental challenge. This study explores the sources, estimation techniques, and mitigation strategies for methane emissions from landfills in the Dhaka landfills, focusing on Matuail and Aminbazar landfills. The study also discusses energy recovery through anaerobic digestion and incineration.

Chapter 1

Introduction

Methane emissions from landfills represent a significant environmental concern due to their impact on climate change. As the third largest anthropogenic source of methane, landfills contribute approximately 11% of global methane emissions, equating to nearly 799 million metric tons of CO₂ equivalent (MMTCO₂E) as of 2010 [1]. This emission is primarily due to the anaerobic decomposition of organic waste in municipal solid waste (MSW) landfills. In 2020, the global generation of waste reached 2.2 billion tons, with about 40% deposited in urban landfills. The biodegradable materials in these landfills are significant contributors to methane production. Methane emissions from landfills are estimated to be equivalent to 900 million tons of CO₂ over a 100-year horizon [1]. This gas is approximately 100 times more potent than carbon dioxide over a 28-year period.

Dhaka has two landfills - Matuail and Aminbazar. The Matuail Sanitary Landfill in Dhaka is a significant source of methane emissions, releasing approximately 4,000 kilograms (4 tonnes) of methane per hour. This amount is equivalent to the emissions produced by about 190,000 traditional cars operating simultaneously [2]. The landfill processes around 2,500 tons of waste daily, contributing to a substantial environmental impact. The Aminbazar landfill receives around 3,200 tons of waste daily from various wards in the city. Leachate from the landfill contaminates local water bodies, making them unsuitable for fishing and irrigation, thereby impacting the economic activities of approximately 22,000 residents who rely on agriculture and fishing [3].

The decomposition of organic waste in landfills occurs in four distinct phases, each characterized by changes in gas composition and production rates. In this initial phase, aerobic bacteria break down complex organic materials in the presence of oxygen, generating carbon dioxide as the primary byproduct. This phase continues until the oxygen supply is exhausted. Once aerobic conditions cease, anaerobic bacteria begin to thrive, converting products from the first

phase into volatile fatty acids, alcohols, and gases like carbon dioxide and hydrogen. This phase typically leads to an increase in the generation of organic acids, which serve as substrates for methane-producing bacteria in subsequent phases [4].

Mitigating and utilizing methane emitted from landfills and converting it into energy can reduce greenhouse gas levels while simultaneously generating renewable energy, thus contributing to environmental sustainability. Additionally, this approach offers economic advantages by creating revenue streams and employment opportunities in sectors related to renewable energy and waste management. It also enhances public health by reducing pollutants associated with methane emissions and aligns with environmental regulations, supporting sustainable urban and industrial practices.

Both developed and developing nations are increasingly recognizing the importance of managing landfills, particularly in capturing methane emissions. Some key methods for landfill waste management are compaction, incineration, anaerobic digestion, and leachate collection and treatment [5]. Compaction involves compressing waste to reduce its volume, which helps minimize landfill space requirements and enhances stability. Incineration is a thermal treatment process where waste is burned at high temperatures, converting it into ash, gases, and heat [5,6]. This method significantly reduces the volume of waste, often by up to 90%, and the resulting heat can be harnessed to produce electricity or heat buildings.

Anaerobic digestion is a biological process that breaks down organic waste, such as food scraps and green waste, in the absence of oxygen. This process occurs in a closed anaerobic digester and produces biogas, a renewable energy source composed mainly of methane. The biogas can be used to generate electricity, and heat, or even be purified and injected into the gas grid for fuel. The process also yields a nutrient-rich digestate, which can be used as a fertilizer, promoting a circular approach to waste management by turning waste into valuable by-products [7]. Leachate collection and treatment systems are essential in landfills to manage the liquid produced when rainwater percolates through waste, dissolving various pollutants [8].

Another way is collecting raw methane gas from existing open landfills. A landfill gas (LFG) collection system typically involves drilling vertical wells into the landfill waste, connected by a network of pipes to collect and transport gas to a central location for treatment or use. The system often includes condensate traps, valves, and blowers to manage vacuum pressure and gas flow [9].

In some studies, the amount of methane emission from landfills of Dhaka is completed [10]. Along with that other models are available for assuming methane emission from landfills [11, 12]. From those models, how much energy can be extracted is calculated for various cities [7, 13, 14]. **In this study we will review a few methods of using landfill waste in Dhaka city, calculate how much energy can be extracted using those models and will estimate how much positive environmental impact can be achieved.**

Chapter 2

Methane Emission Modeling and Estimation

2.1 Landfill Gas (LFG) Composition and Potential

LFG is formed through the biodegradation of solid waste in landfills and typically comprises: 50% methane (CH₄), 45% carbon dioxide (CO₂) and 5% other components (e.g., nitrogen compounds, hydrogen sulphide, NMOCs) [15]. Despite its environmental impact, LFG's high methane content makes it a viable renewable energy source for energy and raw material production.

2.1.1 Methane Estimation

The feasibility and planning of LFG energy recovery projects depend on accurately estimating the amount of methane generated. This can be achieved through:

1. Land Measurements
2. Experimental Processes
3. Mathematical Modelling Approaches

2.2 Modelling Approaches

Various models, including Land GEM (developed by USEPA), IPCC (ZODM, FODM), and others, are widely used to estimate LFG emissions. Models based on zero-order decomposition

are less accurate, and second-order models are less preferred due to parameter determination challenges [16, 17].

Land GEM employs a first-order decomposition equation and requires site-specific parameters such as:

- Methane production rate (K),
- Methane production capacity (L_o),
- Landfill size and amount of MSW dumped per year.

2.3 Calculations

For estimating methane emissions from landfills in Bangladesh, Land GEM V-3.02 is a suitable model. The necessary procedure for methane emission calculation using the Land GEM V-3.02 model is as follows:

$$Q_{CH_4} = \sum_{i=0}^n \sum_{j=0.01}^n K \cdot L_o \cdot \frac{M_i}{10} \cdot e^{k \cdot j}$$

where:

- Q_{CH_4} : Amount of methane emitted (ton/year),
- i : Time increment (1 year),
- j : Time increment (0.10 year),
- K : Methane production rate,
- L_o : Methane production capacity, the total methane gas potential released from one ton of waste,
- M_i : Amount of MSW generated per year.

2.4 Methane Production Capacity (L_o)

In the Land GEM model, five default L_o values have been developed under the Clean Air Act (CAA) and the Compilation of Air Pollutant Emissions factors by the Environmental Protection Agency (EPA). However, these default values are based on waste compositions typical of the United States and may not be suitable for other regions. Therefore, L_o must be calculated specifically for each site to ensure accurate estimations [?, ?].

Methods for Estimating L_o

IPCC Methodology: A more realistic approach that considers practical factors. The L_o value is calculated using the equation:

$$L_o = MCF \cdot DOC \cdot DOC_f \cdot F \cdot \frac{16}{12}$$

where:

- L_o : Potential methane generation capacity (kg/ton),
- MCF : Methane correction factor (default = 1.0),
- DOC : Degradable organic carbon (kg/ton),
- DOC_f : Fraction of assimilated DOC (default = 0.77),
- F : Fraction of methane in LFG (default = 0.5),
- 16/12: Stoichiometric factor for converting carbon to methane.

For our calculation:

$$K = 3.2 \times 10^{-5} \cdot 1930.2 + 0.01 = 0.071766$$

$$L_o = 170 \text{ m}^3/\text{ton (Default CAA)}$$

$$M_i = 1540300 \text{ ton/year.}$$

Year	K	L_o (m ³ /ton)	M_i /day (ton)	j	M_i /Year (ton)	CH ₄ Emission (ton)
2024	0.071766	170	4220	0	1,540,300	1,879,210
2027				3		2,330,659
2030				6		2,890,559
2033				9		3,584,967
2036				12		4,446,194
2039				15		5,514,316
2042				18		6,839,037
2045				21		8,481,998

Table 2.1: Methane Emission Estimation Over Time

Chapter 3

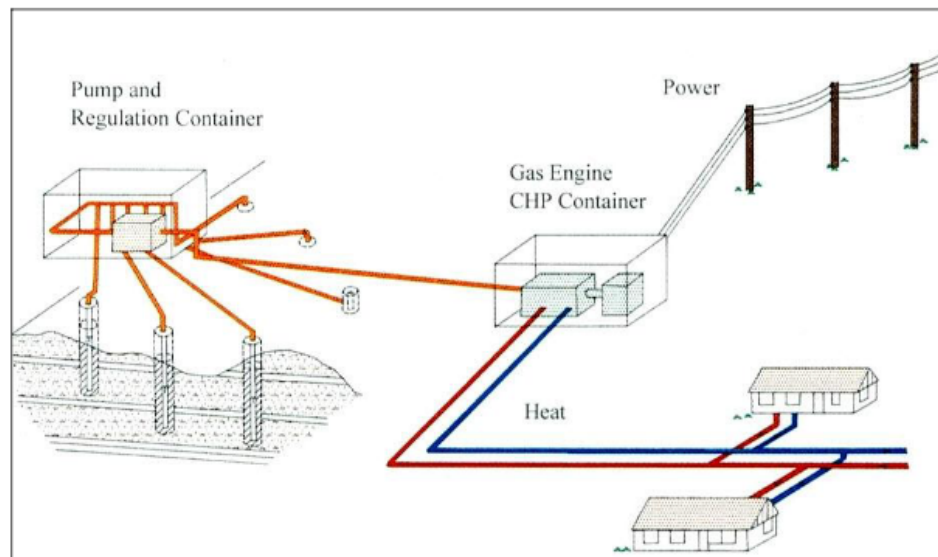
Mitigation and Utilization Methods for Landfill Gas (LFG)

3.1 Landfill Gas (LFG) Extraction Systems and Flaring Overview

Landfill gas (LFG) extraction systems are crucial for managing gases produced during the decomposition of organic waste in landfills. These systems can include vertical perforated pipes, horizontal perforated pipes, ditches, or even impermeable membranes covering the landfill. Among these methods, vertical perforated pipes are the most commonly used due to their simplicity and compatibility with established landfills. Vertical extraction wells are created by drilling holes 50–100 centimeters (cm) in diameter with an auger, then inserting a perforated polyethylene pipe of 10–15 cm diameter into the hole. Gravel is filled around the pipe to enhance gas collection. These wells are typically spaced 40–80 meters apart, depending on landfill depth, to maximize efficiency. However, operational challenges such as well flooding can compromise the functionality of vertical wells. When water accumulates, the gas cannot flow properly. To mitigate this, pipes must be designed to allow pumps to be lowered for water removal. Additionally, condensate traps should be installed at critical points to prevent blockages that could reduce the system's effectiveness.

Horizontal gas extraction systems provide an alternative, especially when installed during landfill filling. These systems use perforated pipes embedded in gravel-filled trenches. The trenches must be sloped to ensure that water and leachate drain into a leachate system or are removed by pumps integrated into the gas system. Typically, trenches are spaced 30–60 meters apart horizontally and 10–25 meters apart vertically. For optimal functionality, the upper layer of the pipes must be at least 3–4 meters below the landfill surface to prevent air infiltration, which can dilute the LFG and reduce its quality. Horizontal systems have the advantage of

beginning gas extraction earlier in the landfill lifecycle, often capturing gas during the peak production phase. However, they are prone to leachate entering the pipes, requiring a robust drainage system to maintain effectiveness.



Source: ©Willumsen, 2009.

Figure 3.1: LFG Recovery Plant System.

In landfills with completed cells, vertical and horizontal systems generally yield similar amounts of LFG. However, horizontal systems offer the advantage of starting gas recovery earlier, potentially increasing the total volume of LFG extracted over time. Vertical systems, by contrast, are easier to install after a landfill has been closed and filled, making them a more practical choice for retrofitting existing sites. Some landfills use impermeable membranes to cover the surface, which can recover up to 90% of LFG. While effective, this method is expensive and reduces moisture content within the waste, which can lower gas production. To counter this, water is often injected beneath the membrane to maintain optimal moisture levels, but achieving uniform water distribution is challenging.

LFG extraction requires the use of pumps or compressors to create a vacuum that pulls the gas from the landfill. Normal operating vacuums range from 20 to 100 millibars at the wellhead. The choice of equipment depends on site-specific needs, particularly the pressure required for gas transport and the inlet pressure for the gas combustion or utilization system. Radial blowers are commonly used because they are simple and economical. Rotary blowers, while more expensive, are reliable and capable of maintaining consistent vacuum and pressure, especially when equipped with frequency regulators. For long transmission pipes or high-pressure applications, screw compressors are preferred. Though expensive, these compressors are highly reliable and have a long operational lifespan.

The extracted gas is typically transported through a network of pipes. Individual wells can be

connected to a central collection pipe, which is positioned to optimize gas flow across the landfill. However, this design can make it difficult to regulate gas quality and quantity or to locate leaks when all wells feed into a single system. A more efficient approach involves connecting each well directly to a pump and regulation house, which allows for better control and reduced operational costs.

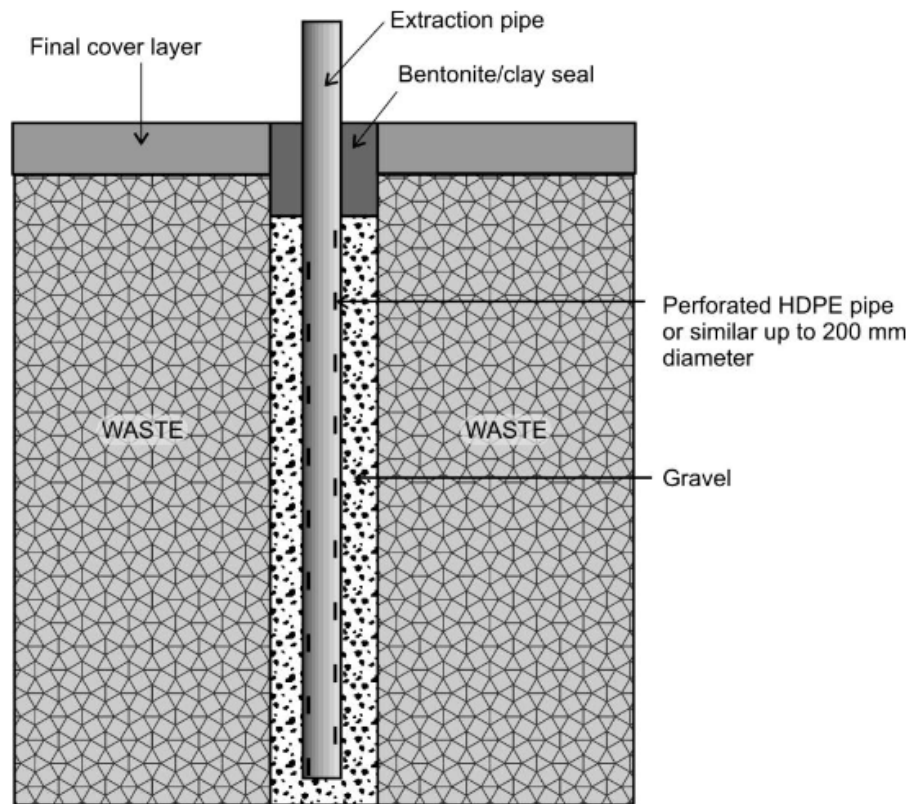


Figure 3.2: Vertical LFG Extraction Well

When using LFG for energy purposes is not feasible, the gas is flared to mitigate environmental and safety risks. Flaring reduces methane emissions, minimizes odors, and lowers the risk of fires or explosions. Two types of flares are commonly used: open and enclosed. Open flares mix LFG with air at the top of the burner, where the mixture is ignited. However, poor mixing and variable flame temperatures result in incomplete combustion and low efficiency, making open flares unsuitable for long-term use. They are typically employed during test periods, start-ups, or temporary interruptions in energy utilization systems. Open flares have an efficiency of around 50

Enclosed flares, on the other hand, are designed for more controlled and efficient combustion. They consist of one or more burners enclosed in a cylindrical structure lined with refractory material. This design prevents quenching and promotes uniform burning, achieving destruction and removal efficiencies of 98–99.5%. Enclosed flares are commonly operated at temperatures of 1,000°C or higher with a minimum retention time of 0.3 seconds, ensuring compliance with stringent emission standards. For Clean Development Mechanism (CDM) projects, specific

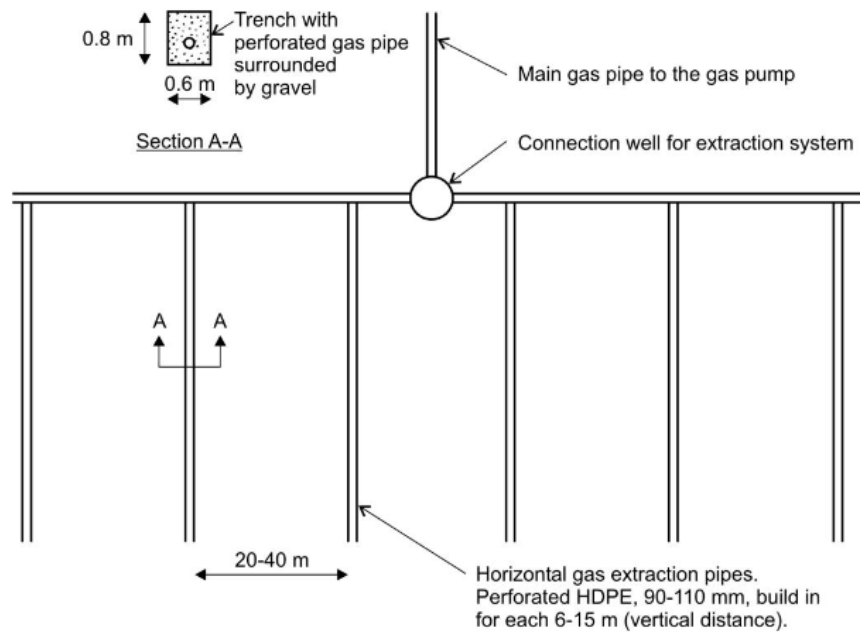


Figure 3.3: Horizontal LFG Extraction System

requirements apply to flare efficiency. Open flares are assigned a default efficiency of 50%, while enclosed flares are allowed a default value of 90%, provided the flare temperature remains above 500°C. Enclosed flares are preferred in such projects due to their higher efficiency and ability to meet stricter environmental standards.

In conclusion, LFG extraction systems play a critical role in landfill management, offering environmental benefits by reducing methane emissions and providing opportunities for energy recovery. Vertical and horizontal extraction systems each have distinct advantages and challenges, and their selection depends on site-specific conditions. When energy recovery is not an option, flaring provides an effective alternative to manage LFG responsibly.

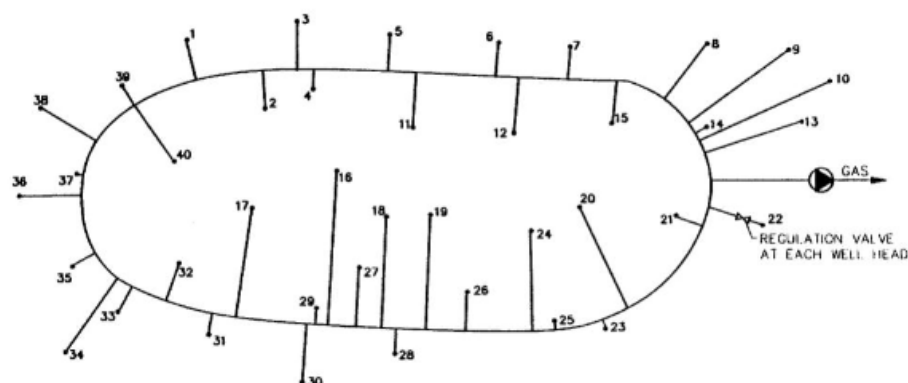


Figure 3.4: Extraction System with Each Well Connected to a Main Collection Pipe

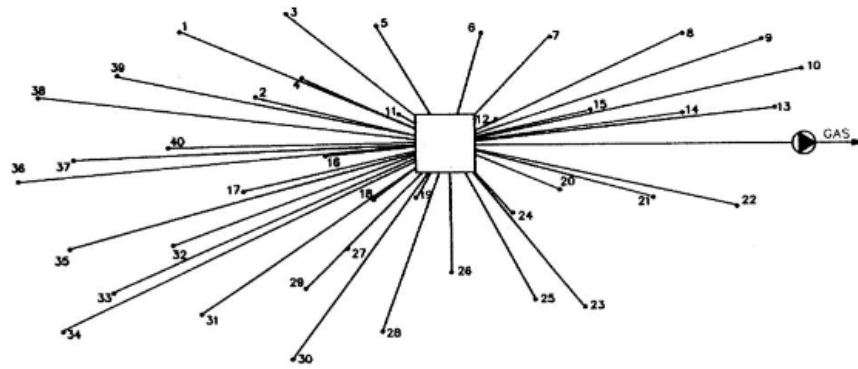


Figure 3.5: Extraction System with Each Well Connected to a Pump and Regulation House

3.2 Anaerobic Digester

Energy can be recovered from the organic fraction of waste (biodegradable as well as non-biodegradable) through two methods as follows:

1. **Thermo-chemical conversion:** This process entails thermal decomposition of organic matter to produce either heat energy or fuel oil or gas; Incineration and Pyrolysis/ Gasification
2. **Bio-chemical conversion:** This process is based on enzymatic decomposition of organic matter by microbial action to produce methane gas or alcohol. Anaerobic Digestion, also referred to as Biomethanation.

Fundamentally, the anaerobic digestion process can be divided into three stages with three distinct physiological groups of micro-organisms: [15]

- **Stage I:** It involves fermentative bacteria, which include anaerobic and facultative micro-organisms. Complex organic materials, carbohydrates, proteins, and lipids are hydrolyzed and fermented into fatty acids, alcohol, carbon dioxide, hydrogen, ammonia, and sulfides.
- **Stage II:** In this stage, the acetogenic bacteria consume these primary products and produce hydrogen, carbon dioxide, and acetic acid.
- **Stage III:** It utilizes two distinct types of methanogenic bacteria. The first reduces carbon dioxide to methane, and the second decarboxylates acetic acid to methane and carbon dioxide. Factors that influence the Anaerobic Digestion process include temperature, pH (Hydrogen Ion Concentration), nutrient concentration, loading rate, toxic compounds, and mixing.

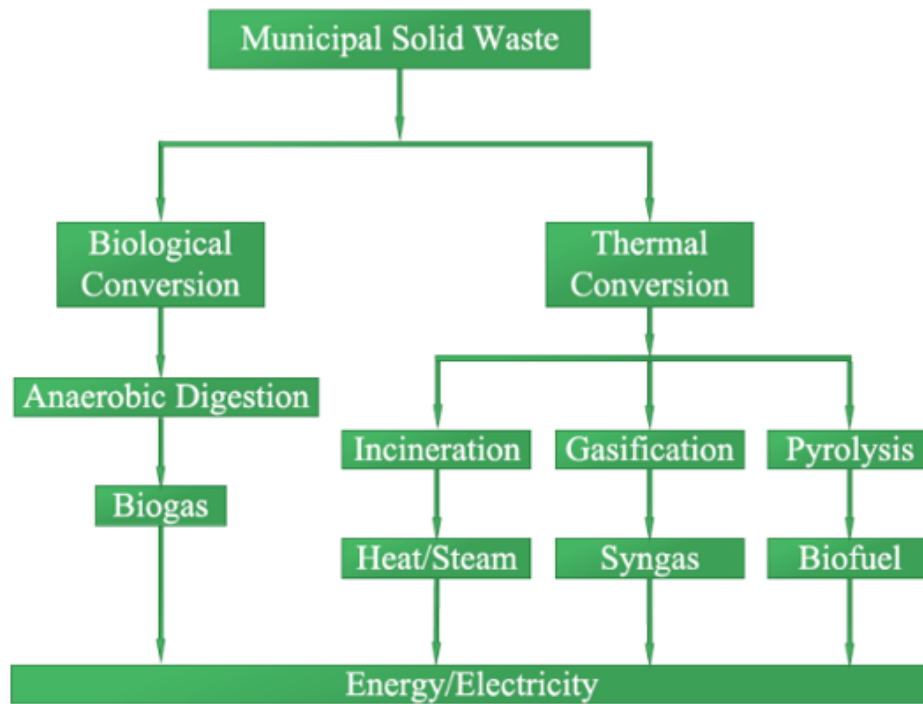


Figure 3.6: Anaerobic Digestion Process

3.2.1 Anaerobic Digestion of Aminbazar and Matuail Landfills

Aminbazar Landfill

Inputs:

- Total waste quantity: 4220 tonnes
- Total Organic/Volatile Solids:

$$\text{VS} = 50\% \text{ of MSW} = 0.5 \times 4220 = 2110 \text{ tonnes}$$

- Organic bio-degradable fraction:

$$66\% \text{ of VS} = 0.66 \times 2110 = 1392.6 \text{ tonnes}$$

- Typical digestion efficiency: 60% [18]

Calculations:

- Typical biogas yield:

$$B(\text{m}^3) = 0.80 \text{ m}^3/\text{kg of VS destroyed} = 0.80 \times 0.60 \times 1392.6 \times 1000 = 668448 \text{ m}^3$$

- Calorific Value of biogas: 5000 kcal/m³ [19]
- Energy recovery potential:

$$\text{Energy (kWh)} = \frac{B \times 5000}{860} = \frac{668448 \times 5000}{860} = 3886325.6 \text{ kWh}$$

- Power generation potential:

$$\text{Power (kW)} = \frac{\text{Energy}}{24} = \frac{3886325.6}{24} = 161930.2 \text{ kW}$$

- Typical Conversion Efficiency = 30% [18, 20]
- Net power generation potential:

$$\text{Net Power (kW)} = 0.3 \times 161930.2 = 48579 \text{ kW} \approx 48.5 \text{ MW}$$

Matuail Landfill

Inputs:

- Total waste quantity: 3257 tonnes
- Total Organic/Volatile Solids:

$$\text{VS} = 50\% \text{ of MSW} = 0.5 \times 3257 = 1628.5 \text{ tonnes}$$

- Organic bio-degradable fraction:

$$66\% \text{ of VS} = 0.66 \times 1628.5 = 1074.8 \text{ tonnes}$$

- Typical digestion efficiency: 60%

Calculations:

- Typical biogas yield:

$$B(\text{m}^3) = 0.80 \text{ m}^3/\text{kg of VS destroyed} = 0.80 \times 0.60 \times 1074.8 \times 1000 = 515908.8 \text{ m}^3$$

- Calorific Value of biogas: 5000 kcal/m³ [19]

- Energy recovery potential:

$$\text{Energy (kWh)} = \frac{B \times 5000}{860} = \frac{515908.8 \times 5000}{860} = 2999469.7 \text{ kWh}$$

- Power generation potential:

$$\text{Power (kW)} = \frac{\text{Energy}}{24} = \frac{2999469.7}{24} = 124977.9 \text{ kW}$$

- Net power generation potential:

$$\text{Net Power (kW)} = 0.3 \times 124977.9 = 37493.4 \text{ kW} \approx 37.4 \text{ MW}$$

Table 3.1: Comparison of Anaerobic Digestion Parameters for Aminbazar and Matuail Landfills

Parameter	Aminbazar	Matuail
Total Waste Quantity (tonnes)	4220	3257
Total Organic/Volatile Solids (VS) (tonnes)	2110	1628.5
Organic Bio-degradable Fraction (tonnes)	1392.6	1074.8
Digestion Efficiency (%)	60%	60%
Bio-gas Yield (m ³)	668,448,000	515,908,800
Calorific Value of Biogas (kcal/m ³)	5000	5000
Energy Recovery Potential (kWh)	3,886,325.60	2,999,469.70
Power Generation Potential (kW)	161,930.20	124,977.90
Typical Conversion Efficiency (%)	30%	30%
Net Power Generation Potential (kW)	48,579	37,493
Net Power Generation Potential (MW)	48.5 MW	37.4 MW

3.3 Incineration

Incinerators are used to burn solid or liquid waste as a part of urban waste management strategy and to extract the energy stored within the waste. Incinerating waste is a cost-effective way to reduce the volume of landfill waste, eliminate toxic compounds and prevent methane emission. Methane is produced from landfills when organic solid waste is allowed to sit in mounds which creates an environment absent of oxygen. Installing incinerators at landfills can prevent buildup of waste and methane emission.

The construction of incinerators is very similar to that of coal furnaces as both handle solid fuel. Solid waste is entered into a combustion chamber and the process is started using supplemental

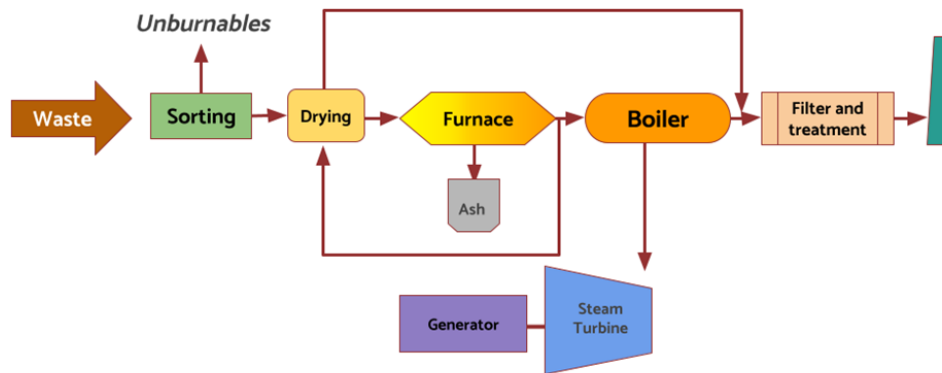


Figure 3.7: Enter Caption

fuel. Once combustion is ongoing the use of supplemental fuel may be stopped depending on the composition and moisture content of the waste. The resulting combustion gases are then used in boilers to generate steam and drive a turbine. The flue gas has to be treated and then released into the atmosphere. As waste is not a good quality fuel significant amount of ash is generated which also has to be treated before disposal.

The construction and working principle of a few types of waste incinerators are briefly described as follows:

- **Rotary Kiln:** A large rotating drum is set at an angle, and solid waste is entered at the top and burned inside. The rotation keeps the fuel moving, allowing for better combustion. The ash flows down out of the kiln, and the flue gas is vented out.
- **Fluidized Bed:** In this method, air is blown through a bed of inert material such as sand, and the fuel is poured from above. This results in the fuel mixing with air properly, leading to more efficient combustion. This type of incinerator is mostly used to burn liquid sludge from sewage treatment plants.
- **Moving Grate:** This is the type most commonly used to incinerate solid municipal waste. Waste is dumped onto one side of a moving grate system, which then burns the waste as it moves through the furnace. Air is passed through the grate. The remaining ash is dumped on the other side of the furnace.

As Bangladesh has a subtropical climate with heavy rainfall during monsoon season, the biggest hurdle in extracting energy through incineration is moisture. Waste that is wet results in lower calorific value as a lot of thermal energy is wasted in vaporizing liquid water. This reduces overall efficiency of the plant. A way to reduce moisture content would be to place the waste in an aerobic environment for some time and allow the natural heat from microbial

activity to drive out the moisture. Another method would be to use a portion of the flue gas generated from the incinerator to dry the waste.

In Dhaka about 7500 tons of solid waste is generated every day. [21] Of that 4220 tons is dumped at Amin Bazar landfill, 3256 tons is dumped at Matuail landfill. [22] All of this waste is left out in the open in large mounds with little to no further processing. This leads to significant pollution and greenhouse gas emission. Using an incinerator would cut down the volume of waste stored, process the waste into a more inert and manageable form and result is considerable energy generation. In order to calculate the energy generation potential through incineration the composition and calorific value of the waste is determined and listed in Table 3.2. The standard calorific value is listed at 25% moisture content. Calorific value is adjusted to 70% moisture content which is found to be the average moisture content for waste in Bangladesh. [23]

Table 3.2: Calorific Value and Composition of Different Waste Types at 70% Moisture Content

Waste Type	Composition (%)	Calorific value (kJ/kg)	Calorific value at 70% Moisture content (kJ/kg)
Paper	12.23	11407.22	10390.22
Organic	70.15	3168.44	2151.44
Wood	4.2	15105.5586	13975.5586
Plastic and glass	4	13342.56	12325.56

Assuming an overall plant efficiency of 20%, [4] the power generation capacity of the incinerator is calculated for each landfill site and shown in Table 3.3.

Table 3.3: Energy Input and Power Output at Different Sites

Site	Amount (tons/day)	Energy Input (MJ/day)	Power Output (MW)
Amin Bazar (DNCC)	4220	1.629×10^7	37.71
Matuail (DSCC)	3256	1.257×10^7	29.09

References

- [1] N. J. Themelis and N. J. Themelis, “How much methane is generated by the global land-filling of urban wastes?,” 4 2023.
- [2] M. A. Amin, “Matuail landfill emits 4 tonnes of methane every hour: “Unbelievable” say experts,” 4 2021.
- [3] B. Devnath, “Aminbazar, the landfill that ruined lives,” 3 2020.
- [4] C. Bond, “Why capturing methane is so difficult,” 2 2024.
- [5] W. Afandi, “Types of Waste Management: Strategies for a Sustainable future,” 6 2024.
- [6] P. E. Escamilla-García, R. H. Camarillo-López, R. Carrasco-Hernández, E. Fernández-Rodríguez, and J. M. Legal-Hernández, “Technical and economic analysis of energy generation from waste incineration in Mexico,” *Energy Strategy Reviews*, vol. 31, p. 100542, 8 2020.
- [7] A. H. Khan, E. A. López-Maldonado, S. S. Alam, N. A. Khan, J. R. L. López, P. F. M. Herrera, A. Abutaleb, S. Ahmed, and L. Singh, “Municipal solid waste generation and the current state of waste-to-energy potential: State of art review,” *Energy Conversion and Management*, vol. 267, p. 115905, 6 2022.
- [8] R. H. Hredoy, M. A. B. Siddique, M. A. Akbor, M. A. A. Shaikh, and M. M. Rahman, “Impacts of landfill leachate on the surrounding environment: A case study on Amin Bazar Landfill, Dhaka (Bangladesh),” *Soil Systems*, vol. 6, p. 90, 11 2022.
- [9] X. Chai, D. J. Tonjes, and D. Mahajan, “Methane emissions as energy reservoir: Context, scope, causes and mitigation strategies,” *Progress in Energy and Combustion Science*, vol. 56, pp. 33–70, 6 2016.
- [10] M. Toha and M. M. Rahman, “Estimation and prediction of methane gas generation from landfill sites in Dhaka city, Bangladesh,” *Case Studies in Chemical and Environmental Engineering*, vol. 7, p. 100302, 1 2023.

- [11] S. Kumar, N. Nimchuk, R. Kumar, J. Zietsman, T. Ramani, C. Spiegelman, and M. Kenney, "Specific model for the estimation of methane emission from municipal solid waste landfills in India," *Bioresource Technology*, vol. 216, pp. 981–987, 6 2016.
- [12] M. Chakraborty, C. Sharma, J. Pandey, N. Singh, and P. K. Gupta, "Methane emission estimation from landfills in Delhi: A comparative assessment of different methodologies," *Atmospheric Environment*, vol. 45, pp. 7135–7142, 9 2011.
- [13] A. Albalade-Ramírez, A. Padilla-Rivera, J. F. Rueda-Avellaneda, B. N. López-Hernández, J. J. Cano-Gómez, and P. Rivas-García, "Mapping the Sustainability of Waste-to-Energy Processes for food loss and waste in Mexico—Part 1: Energy Feasibility Study," *Sustainability*, vol. 16, p. 6111, 7 2024.
- [14] M. Gollapalli and S. H. Kota, "Methane emissions from a landfill in north-east India: Performance of various landfill gas emission models," *Environmental Pollution*, vol. 234, pp. 174–180, 11 2017.
- [15] N. J. Themelis and P. A. Ulloa, "Methane generation in landfills," *Renewable Energy*, vol. 32, pp. 1243–1257, 8 2006.
- [16] H. R. Amini, D. R. Reinhart, and K. R. Mackie, "Determination of first-order landfill gas modeling parameters and uncertainties," *Waste Management*, vol. 32, pp. 305–316, 10 2011.
- [17] Q. Aguilar-Virgen, P. Taboada-González, S. Ojeda-Benítez, and S. Cruz-Sotelo, "Power generation with biogas from municipal solid waste: Prediction of gas generation with in situ parameters," *Renewable and Sustainable Energy Reviews*, vol. 30, pp. 412–419, 11 2013.
- [18] S. Jeyapriya and M. Saseetharan, "Energy recovery from municipal solid waste in an anaerobic reactor," *Journal of Environmental Science and Engineering*, vol. 50, no. 3, pp. 235–238, 2008.
- [19] U. D. of Energy, *The Biogas Handbook*. 1 2013.
- [20] "Landfill Gas Energy Project Data — US EPA," 9 2024.
- [21] M. J. Islam, "Dhaka landfills in disarray," 8 2021.
- [22] "Waste Report 2018-2019," 2019.
- [23] Admin and Admin, "FEASIBILITY STUDY WASTE TO ENERGY POWER PLANT FOR NAWABGANJ, KERANIGANJ, AND DOHAR," 10 2024.

Generated using Undegraduate Thesis L^AT_EX Template, Version 1.4. Department of
Mechanical Engineering, Bangladesh University of Engineering and Technology,
Dhaka, Bangladesh.

This thesis was generated on Monday 9th December, 2024 at 7:59am.